01 / PLANNED

Manufacture the stand from CAD

1. **CAD package complete**

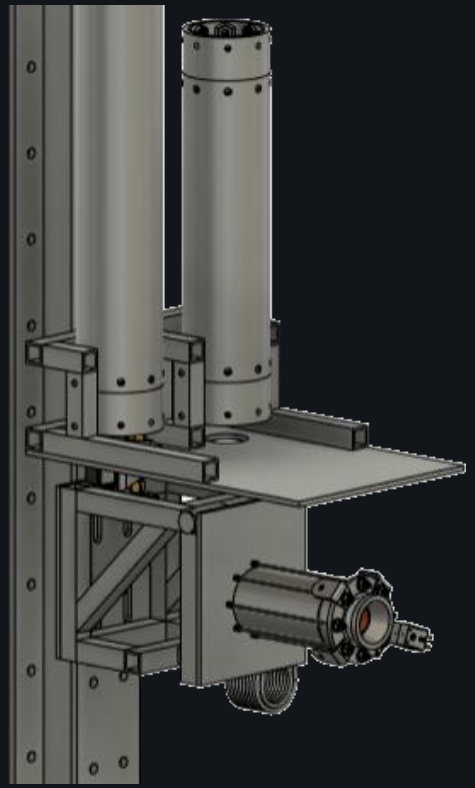
   Tank brackets, engine plate, plumbing routing, load cell hinge all modeled.

2. **Build to print**

   Cut, weld and assemble the structure, then seat both tanks in their brackets.

3. **Instrument for hot fire**

   Load cell in the thrust path to capture real engine performance data.



CAD — as designed

The stand got built — three catches

1 Weld tolerancing

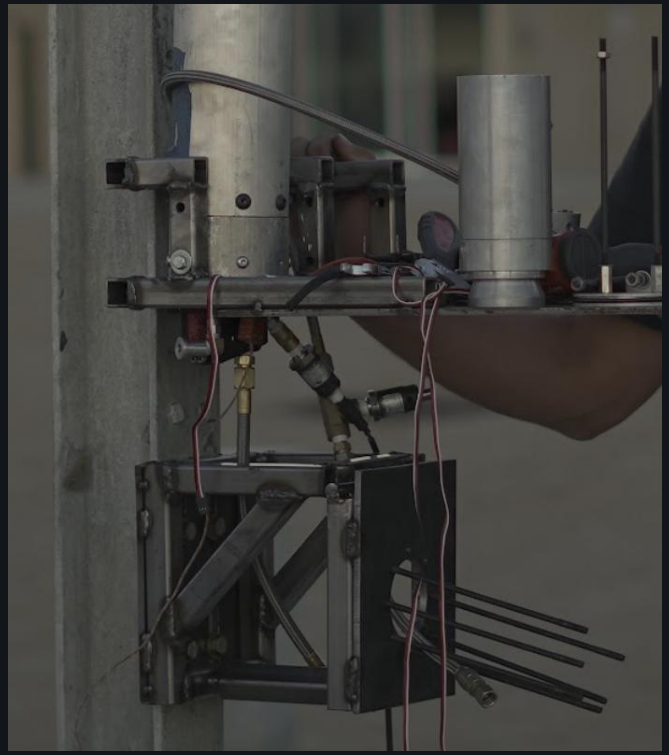
Bracket welds were not taken into account, so the tanks did not drop cleanly into their mounts.

2 Load cell not purchased

A solid support went in its place. The stand holds, but we get no thrust data from it.

3 Plumbing access

Fittings were difficult to access with the configuration of the blast plate having the tank in line with it and not having lots of space below blast plate.



As built — solid support where the load cell belongs

03 / NEXT

Redesign into a modular test stand

1

Sensor integration

Add the load cell into the test stand and think of better ways for all sensors to integrate into the test stand.

2

Plumbing access

Space apart the structure more to give better access into plumbing and easier tank mounting.

3

Modularity

Create a modular setup to accept any tank or engine design.